

1. Administrative & Study Identification

Full Study Title: A Phase Ib/II Open-label, Multi-center Study of AM0959 With Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) in Combination With an Androgen Receptor Pathway Inhibitor (ARPI) in Adult Participants With PSMA-positive Metastatic Castration Resistant Prostate Cancer (mCRPC) **Short Title/Acronym:** AM0959 and AAA617 for PSMA-positive mCRPC **Protocol Number:** CAM0959A12103 **NCT Number:** NCT07226986 **Posting ID:** 1079774794003581831 **Trial Status:** NOT_YET_RECRUITING **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** AM0959 + Lutetium (177Lu) vipivotide tetraxetan (AAA617 / Pluvicto) + ARPI (abiraterone or enzalutamide) **Phase of Development:** Phase Ib/II **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * **Phase Ib:** To evaluate the safety, tolerability, and pharmacokinetics (PK) to determine the recommended dose for expansion (RDE) of AM0959 when given in combination with AAA617 and ARPI in participants with mCRPC who have failed one prior ARPI. * **Phase II:** To evaluate the preliminary efficacy of AM0959 in combination with AAA617 and ARPI in participants with mCRPC who have failed one prior ARPI, but who have not yet been exposed to taxane treatment.

Secondary Objectives: To assess radiographic progression-free survival (rPFS), PSA90 response, PSA50 response, time to first symptomatic skeletal event (TTSSE), and change from baseline in FACT-P PCS.

2.2 Background & Rationale To address the need for improved treatments in metastatic castration-resistant prostate cancer (mCRPC) by evaluating the potential synergistic effect and clinical efficacy of combining the targeted investigational agent AM0959 with an approved radioligand therapy (177Lu-PSMA-617) and an androgen receptor pathway inhibitor (ARPI).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Phase Ib dose escalation) / Active experimental cohorts (Phase II expansion) **Blinding:** Open-label **Randomization:** N/A for Phase Ib (sequential dose escalation); Randomized for Phase II

3.2 Planned Enrollment & Duration Target **\$N\$:** 123 participants (Phase Ib features provisionally 3 dose-level cohorts of 3-6 participants each. Phase II will follow with 25 participants per arm randomized in a 1:1:1 ratio) **Number of Sites:** Multi-center (Global, including Australia) **Estimated Study Start:** 05-Dec-2025 **Estimated Primary Completion:** 10-Jul-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male adults ≥ 18 years of age with an ECOG performance status of 0 to 2. 2. Histologically confirmed adenocarcinoma of the prostate (mCRPC) with documented progressive disease while on treatment with ARPI in mHSPC or earlier setting as their last treatment (and did not progress on more than one ARPI). 3. PSMA-PET positive disease assessed by an approved PSMA imaging agent as determined by central reading rules. 4. Phase Ib: Prior exposure of up to 1 line of taxane-based chemotherapy is permissible. Phase II: Must not have received taxane-based chemotherapy in mCRPC setting (allowed in mHSPC setting). 5. Castration level of testosterone (< 50 ng/dL), and/or use of concomitant ADT.

4.2 Exclusion Criteria 1. Concurrent local (radiation therapy to the prostate with curative intent or other prostate antineoplastic ablative procedures) or systemic (hormonal ablation, chemotherapy, immunotherapy) antineoplastic treatments within 28 days of enrollment/randomization. 2. Prior treatment with any radioligand therapy (RLT) or PSMA-targeted agents (approved or investigational). 3. Participants with a history of CNS metastases unless they have received therapy (surgery, whole brain radiation therapy, stereotactic radiosurgery), are neurologically stable, asymptomatic, and not taking corticosteroids for the purpose of maintaining neurologic integrity. 4. Any other investigational agents within 28 days prior to first dose of any study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants will receive AM0959 BID (days 2-15), combined with AAA617 administered on Day 1 every 6 weeks (for a maximum of 6 cycles), along with continuous ARPI (abiraterone or enzalutamide) starting on Day 1. **Route of Administration:** AM0959 (Oral), AAA617 (Intravenous), ARPI (Oral). **Duration of Treatment:** Up to 6 cycles (each cycle is 6 weeks). Follow-up assessments continue for approximately 45 months.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant androgen deprivation therapy (ADT) to maintain castration levels of testosterone (< 50 ng/dL). **Prohibited:** Other investigational

agents within 28 days prior to the first dose, and concurrent unapproved local/systemic antineoplastic therapies.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, ECOG, PSMA-PET imaging, Baseline PSA & PK Labs | | **Baseline** | Day 1 | Randomization/Enrollment, First Dose of AAA617, ARPI start | | **Interim** | Every 6 Weeks | AAA617 dosing (max 6 cycles), AM0959 BID dosing (Days 2-15 of cycle), Safety Labs, PSA Response, DLT evaluation (Phase Ib) | | **End of Study** | Up to ~45 months | Final rPFS Assessment, TTSSE tracking, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Phase Ib Primary Outcomes: 1. Incidence rate of Dose-limiting toxicities (DLTs) graded by NCI CTCAE version 5.0. 2. Incidence rate of Adverse Events (AEs) and Serious Adverse Events (SAEs) by type, frequency, and severity, graded by NCI CTCAE version 5.0. 3. Number of participants with dose adjustments (reductions, interruption, or permanent discontinuation). **Phase II Primary Outcome:** Preliminary efficacy indicated by disease response metrics.

7.2 Secondary & Exploratory Endpoints * Radiographic progression-free survival (rPFS) (Phase Ib and Phase II) * Prostate Specific Antigen 90 (PSA90) response * Prostate Specific Antigen 50 (PSA50) response (Phase Ib) * Time to first symptomatic skeletal event (TTSSE) * Change from baseline in FACT-P PCS

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via physical exams and labs; specifically evaluating Dose-Limiting Toxicities (DLTs) in Phase Ib using NCI CTCAE v5.0. **Serious Adverse Events (SAEs):** Standard protocol, immediate reporting. **Data Monitoring Committee (DMC):** Dose Escalation Meetings (DEMs) will occur when all participants in a Phase Ib dose level cohort have completed the DLT evaluation period or experienced a DLT prior to the end of the period, to assess safety before advancing.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Phase Ib safety set (all who receive at least one dose); Phase II efficacy set (Intent-to-Treat / randomized participants). **Sample Size Justification:** Phase Ib follows a standard 3 provisional cohort dose-escalation design;

Phase II sample size of 25 participants per arm (randomized 1:1:1) is powered to detect preliminary efficacy signals in PSA and rPFS responses. **Handling of Missing Data:** Standard survival analytic frameworks with data censoring for missing end-of-study endpoints (e.g., rPFS, TTSSE).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard multi-center Phase Ib/II clinical trial monitoring frequency by Novartis. **Audit Trail:** eCRF locking, strict tracking and accountability for investigational drug and radioligand substances (AAA617).

11. Study Identifiers & Governance

Primary ID: CAM0959A12103 **Registry Number:** NCT07226986 **Posting ID:** 1079774794003581831 **Trial Status:** NOT_YET_RECRUITING **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** AM0959 and AAA617 for PSMA-positive mCRPC **Official Regulatory Title:** A Phase Ib/II Open-label, Multi-center Study of AM0959 With Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) in Combination With an Androgen Receptor Pathway Inhibitor (ARPI) in Adult Participants With PSMA-positive Metastatic Castration Resistant Prostate Cancer (mCRPC)

12. Clinical Framework (Oncology/mCRPC Specific)

Primary Condition: PSMA-positive Metastatic Castration Resistant Prostate Cancer (mCRPC) With Prior Exposure to One Prior ARPI Who Are Candidates for Taxane-based Chemotherapy **Preferred UMLS Name:** PSMA-Positive Metastatic Castration-Resistant Prostate Carcinoma **MeSH Code:** D009369 **SNOMED ID:** 263650009 **ATC Code (Abiraterone):** L02BX03 **Diagnosis Threshold:** Serum/plasma PSA progression (e.g., min start value 2.0 ng/mL with confirmed rises) & PSMA-PET positive disease **Medical Methodology:** Targeted Radioligand Therapy + AM0959 + ARPI **Optimization Target:** PSA reduction (PSA50/PSA90) and radiographic progression-free survival (rPFS)

13. Study Procedures & Methodology

Management Pathway: AM0959 + AAA617 given cyclically (every 6 weeks) with continuous ARPI **Education Protocol (HCP):** Site-level protocol and radioactive materials (177Lu) handling/safety training **Education Protocol (Patient):** Informed consent, adverse event tracking, and radiation safety instructions **Quality Audit**

Mechanism: Phase Ib Dose Escalation Meetings (DEMs) and standard efficacy tracking

14. Observation & Follow-Up Schedule

Total Duration: Follow-up assessed up to approximately 45 months
Onsite Visit Cadence: Baseline, Day 1 of each 6-week cycle, and necessary follow-ups
Remote Monitoring Frequency: Based on institutional standard for Phase Ib/II patient safety
Checklist Review Interval: Every 6 weeks (aligned with treatment cycles)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					